

MSSQL Backup and Restore Assignment

Topic

Full Backup, Differential Backup, and Restore in Microsoft SQL Server

Scenario

A telecom company stores important operational data such as:

- Subscribers
- Recharge transactions
- Internet package subscriptions
- Billing records
- Customer service logs

The company must ensure:

- Data protection
- Disaster recovery
- Business continuity
- Backup management
- Recovery operations

The Database Administrator (DBA) is responsible for implementing backup and restore strategies using:

- Full Backup
- Differential Backup
- Database Restore

Students must perform backup and recovery operations for the telecom database.

Database Setup Create Database

```
CREATE DATABASE TelecomBackupDB;
```

```
GO
```

```
USE TelecomBackupDB;
```

GO

Create Subscribers Table

```
CREATE TABLE subscribers (  
    subscriber_id INT PRIMARY KEY IDENTITY(1,1),  
    subscriber_name VARCHAR(100),  
    phone_number VARCHAR(20),  
    package_name VARCHAR(50),  
    balance MONEY,  
    created_at DATETIME DEFAULT GETDATE()  
);  
GO
```

Insert Sample Data

```
INSERT INTO subscribers  
(subscriber_name, phone_number, package_name, balance)  
VALUES  
( 'Aung Aung', '091111111', 'Basic', 5000),  
( 'Su Su', '092222222', 'Premium', 10000),  
( 'Kyaw Kyaw', '093333333', 'Student', 3000);  
GO
```

Assignment Questions

Question 1 — Create Full Backup Requirement

Create a Full Backup of the TelecomBackupDB database.

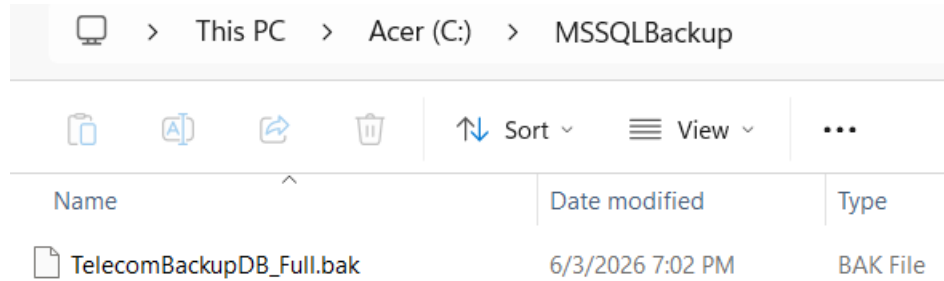
Backup File Name

TelecomBackupDB_Full.bak

Backup Location

C:\MSSQLBackup\

```
25  BACKUP DATABASE TelecomBackupDB
26  TO DISK = 'C:\MSSQLBackup\TelecomBackupDB_Full.bak'
27  WITH INIT, NAME = 'Full Backup of TelecomBackupDB';
```



Name	Date modified	Type
TelecomBackupDB_Full.bak	6/3/2026 7:02 PM	BAK File

A full backup creates a complete copy of the entire database, including all objects and data. This ensures that the database can be fully restored to the exact state at the time of backup.

- File: TelecomBackupDB_Full.bak
- Location: C:\MSSQLBackup\
- Purpose: Provides a baseline backup for disaster recovery.

Question 2 — Insert Additional Data Requirement

Insert at least 3 new subscriber records after the Full Backup operation.

```
29  INSERT INTO subscribers (subscriber_name, phone_number, package_name, balance)
30  VALUES
31  ('Mya Mya', '094444444', 'Premium', 8000),
32  ('Hla Hla', '095555555', 'Basic', 4000),
33  ('Ko Ko', '096666666', 'Student', 2000);
```

Question 3 — Create Differential Backup Requirement

Create a Differential Backup after inserting new subscriber records.

Backup File Name

TelecomBackupDB_Diff.bak

```
35  BACKUP DATABASE TelecomBackupDB
36  TO DISK = 'C:\MSSQLBackup\TelecomBackupDB_Diff.bak'
37  WITH DIFFERENTIAL, INIT, NAME = 'Differential Backup of TelecomBackupDB';
```

> This PC > Acer (C:) > MSSQLBackup		
	Sort ▾	View ▾
Name	Date modified	Type
TelecomBackupDB_Full.bak	6/3/2026 7:02 PM	BAK File
TelecomBackupDB_Diff.bak	6/3/2026 7:04 PM	BAK File

A differential backup only stores changes made since the last full backup. This makes it faster and smaller compared to full backups.

- File: TelecomBackupDB_Diff.bak
- Purpose: Captures newly inserted subscriber records without duplicating unchanged data.

Question 4 — Simulate Data Loss Requirement

Delete all records from the subscribers table to simulate accidental data loss.

39 | `DELETE FROM subscribers;`

SQLQuery3.s...Yu Ya (107))

MSSQL Backup...y Yu Ya (51))

1	<code>SELECT TOP (1000) [subscriber_id]</code>
2	<code>, [subscriber_name]</code>
3	<code>, [phone_number]</code>
4	<code>, [package_name]</code>
5	<code>, [balance]</code>
6	<code>, [created_at]</code>
7	<code>FROM [TelecomBackupDB].[dbo].[subscribers]</code>
8	

100 % ▾ No issues found

Results Messages

subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
---------------	-----------------	--------------	--------------	---------	------------

To test recovery, all records in the subscribers table were deleted. This simulates accidental data loss, a common scenario in real-world operations.

- Purpose: Validates the effectiveness of backup and restore strategies.
-

Question 5 — Restore Full Backup Requirement

Restore the database using the Full Backup file.

Expected Result

Only original records should be restored.

```
41 USE master;
42 ALTER DATABASE TelecomBackupDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
43
44 RESTORE DATABASE TelecomBackupDB
45 FROM DISK = 'C:\MSSQLBackup\TelecomBackupDB_Full.bak'
46 WITH REPLACE;
47
48 ALTER DATABASE TelecomBackupDB SET MULTI_USER;
```

SQLQuery4.s... Yu Ya (56) MSSQL Backup...y Yu Ya (51))

```
1 SELECT TOP (1000) [subscriber_id]
2     , [subscriber_name]
3     , [phone_number]
4     , [package_name]
5     , [balance]
6     , [created_at]
7 FROM [TelecomBackupDB].[dbo].[subscribers]
8
```

100 % No issues found

Results Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	1	Aung Aung	091111111	Basic	5000.00	2026-06-03 19:00:45.090
2	2	Su Su	092222222	Premium	10000.00	2026-06-03 19:00:45.090
3	3	Kyaw Kyaw	093333333	Student	3000.00	2026-06-03 19:00:45.090

Restoring from the full backup reverts the database to its original state, containing only the initial subscriber records.

- Result: Only the three original records are recovered.

Question 6 — Restore Differential Backup

Requirement

Restore the database using:

1. Full Backup
2. Differential Backup

Expected Result

All original and newly inserted records should be restored.

```
50 USE master;
51 ALTER DATABASE TelecomBackupDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
52
53 RESTORE DATABASE TelecomBackupDB
54 FROM DISK = 'C:\MSSQLBackup\TelecomBackupDB_Full.bak'
55 WITH NORECOVERY;
56
57 RESTORE DATABASE TelecomBackupDB
58 FROM DISK = 'C:\MSSQLBackup\TelecomBackupDB_Diff.bak'
59 WITH RECOVERY;
60
61 ALTER DATABASE TelecomBackupDB SET MULTI_USER;
```

SQLQuery5.s... Yu Ya (94)) X MSSQL Backup...y Yu Ya (51))

```
1 SELECT TOP (1000) [subscriber_id]
2     , [subscriber_name]
3     , [phone_number]
4     , [package_name]
5     , [balance]
6     , [created_at]
7 FROM [TelecomBackupDB].[dbo].[subscribers]
8
```

100 % No issues found

Results Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	1	Aung Aung	091111111	Basic	5000.00	2026-06-03 19:00:45.090
2	2	Su Su	092222222	Premium	10000.00	2026-06-03 19:00:45.090
3	3	Kyaw Kyaw	093333333	Student	3000.00	2026-06-03 19:00:45.090
4	4	Mya Mya	094444444	Premium	8000.00	2026-06-03 19:03:37.690
5	5	Hla Hla	095555555	Basic	4000.00	2026-06-03 19:03:37.690
6	6	Ko Ko	096666666	Student	2000.00	2026-06-03 19:03:37.690

Restoring the full backup first, followed by the differential backup, recovers both the original and newly inserted records.

- Result: All six subscriber records are restored.

Question 7 — Verify Recovery

Requirement

Verify that all subscriber records are successfully restored.

```
66 SELECT * FROM subscribers;
67
```

100 % 1 0

Results Messages

	subscriber_id	subscriber_name	phone_number	package_name	balance	created_at
1	1	Aung Aung	0911111111	Basic	5000.00	2026-06-03 19:00:45.090
2	2	Su Su	0922222222	Premium	10000.00	2026-06-03 19:00:45.090
3	3	Kyaw Kyaw	0933333333	Student	3000.00	2026-06-03 19:00:45.090
4	4	Mya Mya	0944444444	Premium	8000.00	2026-06-03 19:03:37.690
5	5	Hla Hla	0955555555	Basic	4000.00	2026-06-03 19:03:37.690
6	6	Ko Ko	0966666666	Student	2000.00	2026-06-03 19:03:37.690

After restoration, a `SELECT * FROM subscribers;` query confirms that all records are present.

- Purpose: Ensures recovery operations were successful.

Question 8 — Backup Verification

Requirement

Write SQL commands to verify backup files.

```
68 RESTORE VERIFYONLY
69 FROM DISK = 'C:\MSSQLBackup\TelecomBackupDB_Full.bak';
70
71 RESTORE VERIFYONLY
72 FROM DISK = 'C:\MSSQLBackup\TelecomBackupDB_Diff.bak';
73
```

100 % 1 0

Messages

The backup set on file 1 is valid.
The backup set on file 1 is valid.

The `RESTORE VERIFYONLY` command checks the integrity of backup files without actually restoring them.

- Purpose: Confirms that backup files are valid and usable.

Question 9 — Backup History Requirement

Display backup history information from system tables.

```

74 SELECT database_name, backup_start_date, backup_finish_date, backup_size, type, physical_device_name
75 FROM msdb.dbo.backupset b
76 JOIN msdb.dbo.backupmediafamily m
77 ON b.media_set_id = m.media_set_id
78 WHERE database_name = 'TelecomBackupDB'
79 ORDER BY backup_finish_date DESC;
80

```

	database_name	backup_start_date	backup_finish_date	backup_size	type	physical_device_name
1	TelecomBackupDB	2026-06-03 19:04:49.000	2026-06-03 19:04:49.000	999424	I	C:\MSSQLBackup\TelecomBackupDB_Diff.bak
2	TelecomBackupDB	2026-06-03 19:02:40.000	2026-06-03 19:02:41.000	4210688	D	C:\MSSQLBackup\TelecomBackupDB_Full.bak

System tables in the msdb database store metadata about backup operations. Querying these tables provides details such as backup type, size, and timestamp.

- Purpose: Helps DBAs track and audit backup activities.

Question 10 — Recovery Model

Requirement

Check the recovery model of the database.

```

81 SELECT name, recovery_model_desc
82 FROM sys.databases
83 WHERE name = 'TelecomBackupDB';
84

```

	name	recovery_model_desc
1	TelecomBackupDB	SIMPLE

The recovery model defines how transactions are logged and how backups can be performed. Common models include:

- **Simple:** Minimal logging, no point-in-time recovery.
 - **Full:** Complete logging, supports point-in-time recovery.
 - **Bulk-Logged:** Optimized for bulk operations.
 - Purpose: Ensures the database is configured for the required level of protection.
-